

0.1 Introduction

First-Person-View (FPV) racing drones require precise and responsive control in order to achieve stable flight behaviour, suppress disturbances such as propeller wash and maintain agility during rapid manoeuvres. The quality of this behaviour strongly depends on the tuning of the flight controller's feedback loops. In particular, the parameters of the angular rate controller determine how quickly and accurately the drone reacts to pilot inputs and external disturbances.

In practice, controller tuning is still predominantly performed manually through iterative trial-and-error procedures. This process requires extensive flight testing and considerable piloting experience, making it both time-consuming and difficult to reproduce. Even though modern flight controller firmware such as Betaflight offers extensive configuration possibilities and detailed flight data logging, efficient and systematic tuning remains a significant challenge.

A previous project introduced a proof-of-concept approach for offline tuning of angular rate controllers using recorded flight data and system identification techniques. Since then, this concept has been further developed beyond the initial proof of concept and forms the basis for the present work.

The objective of this bachelor thesis is to further refine and extend this approach into a modular offline tuning framework for multiple control loops used in Betaflight-based flight controllers. In addition to the already developed angular rate controller tuner, new modules will be introduced for tuning the angle controller, the altitude hold controller, and the position controller. These additions aim to extend the tuning framework beyond the inner rate control loop and provide tools for analysing and tuning higher-level control layers.

The framework will be implemented primarily in MATLAB and Python and will provide tools for analysing flight logs, estimating system dynamics and generating controller parameters without requiring repeated flight tests. The software will be developed as an open-source library and accompanied by comprehensive documentation to ensure accessibility and usability for the FPV community.

By providing a structured, data-driven approach to controller tuning, this project aims to simplify the tuning process, improve flight performance and reduce the reliance on manual trial-and-error methods.

0.2 Goal / Task Description

The goal of this bachelor thesis is to develop a modular offline tuning framework for multiple control loops used in Betaflight-based flight controllers. Newly, the angle controller and the altitude hold controller should be added to the already developed angular rate controller tuner.

If time allows, a position controller tuner should be also implemented as well. The framework should be implemented primarily in MATLAB and Python.

To achieve this goal, the firmware of the flight controller will be modified to add the chirp signal to the input signal of the angle controller, the altitude hold controller and the position controller.

0.3 Specification

0.3.1 Technical Specification

For the implementation of the offline tuning framework, the following technical specifications are defined:

- The framework should be implemented in MATLAB and Python.
- The framework should be modular, allowing for the addition of new tuning modules for different control loops.
- The framework should provide tools for analysing flight logs, estimating system dynamics and generating controller parameters.
- The framework should be developed as an open-source library and accompanied by comprehensive documentation.

0.3.2 Functional Specification

The functional specifications for the offline tuning framework are as follows:

- The framework should be able to process flight logs recorded with Betaflight firmware.
- The framework should be able to estimate the system dynamics of the angle controller, the altitude hold controller and the position controller based on the recorded flight data.
- The framework should be able to generate controller parameters for the angle controller, the altitude hold controller and the position controller based on the estimated system dynamics.
- The framework should provide a user-friendly interface for configuring and running the tuning process.

0.3.3 Boundary Conditions

The following boundary conditions are defined for the offline tuning framework:

- The framework should be compatible with Betaflight firmware version
- The framework should be designed for use with FPV racing drones equipped with Betaflight-based flight controllers.

0.4 Costs

The costs associated with the development of the offline tuning framework are estimated as follows:

- Additional GPS modules for testing the position controller tuner

0.5 Work Packages

The project is divided into several work packages that structure the development and validation of the offline tuning framework.